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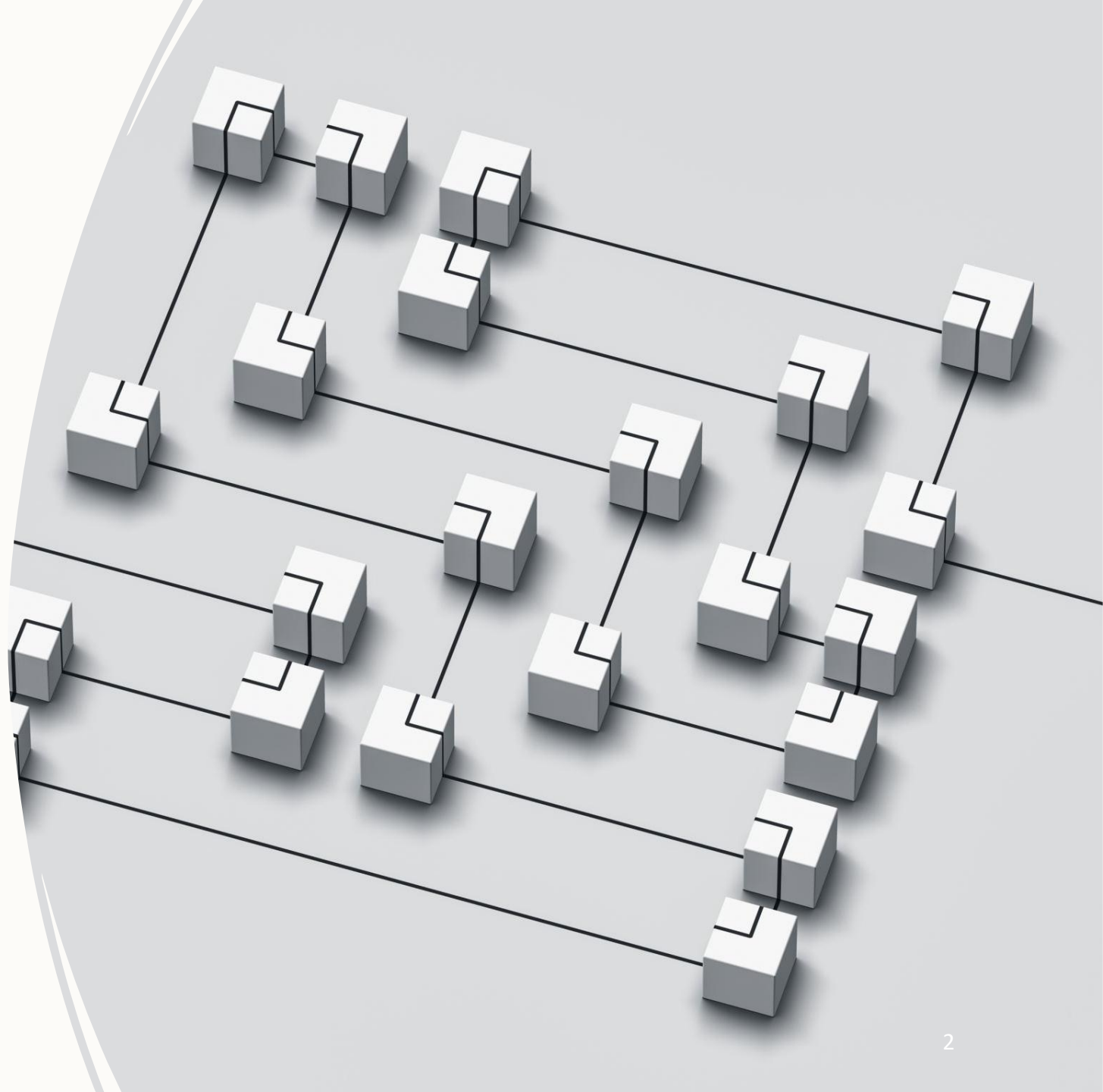
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COMP 3056 Professional Ethics in Computing

Week 7 Privacy

Learning outcomes

- Perspectives on privacy
 - Dimensions of privacy
 - Threats and principles
 - Trade-offs and rights
- Theories of privacy
 - Definitions and views
 - Solove's Taxonomy
- Different Opinions on Privacy





Additional Reading

Books

- Chapter 3 of the book Ethics in a Computing Culture (Brinkman & Sanders, 2013)
- Chapter 2 of Sara Baase (2013), A gift of fire: social, legal, and ethical issues of computing technology.

Articles

- I've Got Nothing to Hide and Other Misunderstandings of Privacy. Daniel J. Solove, San Diego Law Review, Vol. 44, pp. 745—772, 2007.
- Can You Engineer Privacy? Seda Gurses. Communications of the ACM, Vol. 57, No. 8, pp. 20-23, August 2014.
- Security and Privacy of Augmented Reality Systems. Franziska Roesner, Today Oshi Kohno, and David Molnar. Communications of the ACM, Vol. 57, No. 4, pp.88-96, April 2014.

Articles are available on Moodle.





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Perspectives on Privacy

What does privacy mean to different people?

Student assignment – an example

- Professor Blake teaches computer security. In that module, he teaches students how easy it is to intercept e-mail and instant messages. As an assignment, students are required to intercept e-mails and instant messages from the University's network and post them to the class blog.
- Jessica - one of the students on the module – objects to the assignment on the grounds that it constitutes an invasion of privacy. Professor Blake disagreed for the following two reasons...
 - It is very easy to intercept e-mail, so emails cannot be considered private.
 - The email accounts are on University servers, the contents of which are actually public, so reading them is not a privacy violation.
- **Identify important questions that arise from this scenario.**

Student assignment – some questions

- Expectations of privacy?
- Reasonable to expect e-mail to be private?
- Making information public without knowledge?
- Making information accessible?
- Difference between morality, ethical, and privacy issue?
- Sensibility about the content and the authorship?
- Gaining consent eliminates the problem?
- Secrecy, intrusion, control, surveillance: who is to blame?
- Role of computing professionals?

这一页的本质：教你从多角度分析一个“看似只是安全作业”的场景：技术上可行 不等于 伦理上合理。



A summary of privacy risks

- Anything we do in cyberspace is recorded, at least briefly, and linked to our computer or phone, and possibly our name.
- With the huge amount of storage space available, companies, organizations, and governments save huge amounts of data that no one would have imagined saving in the recent past.
- People often are not aware of the collection of information about them and their activities.
- Software is extremely complex. Sometimes businesses, organizations, and website managers do not even know what the software they use collects and stores.⁸
- Leaks happen. The existence of the data presents a risk.
- A collection of many small items of information can give a fairly detailed picture of a person's life.
- Direct association with a person's name is not essential for compromising privacy. Re-identification has become much easier due to the quantity of personal information stored and the power of data search and analysis tools.
- If information is on a public website, people other than those for whom it was intended will find it. It is available to everyone.
- Once information goes on the Internet or into a database, it seems to last forever. People (and automated software) quickly make and distribute copies. It is almost impossible to remove released information from circulation.
- It is extremely likely that data collected for one purpose (such as making a phone call or responding to a search query) will find other uses (such as business planning, tracking, marketing, or criminal investigations).
- The government sometimes requests or demands sensitive personal data held by businesses and organizations.
- We often cannot directly protect information about ourselves. We depend on the businesses and organizations that manage it to protect it from thieves, accidental collection, leaks, and government prying.



Baase (2013), A gift of fire: social, legal, and ethical issues of computing technology, pp. 55-56.



A list of privacy risks & threats

- A. Intentional use or release 有意使用或公开
- B. Unauthorised use or release 未经授权的使用/披露
- C. Inadvertent leakage or careless loss 无意泄露或粗心丢失
- D. Search query data 搜索查询数据
- E. Re-identification of individuals 个人重新识别
- F. Smartphone data 智能手机数据
- G. Massive data storage (i.e., the cloud) 海量存储/云端
- H. Complexity of software 软件复杂性
- I. Surveillance and recognition 监控与识别
- J. Unintended use 非预期用途
- K. Difficult to control or protect (e.g. *autonomous systems such as AI*). 难以控制或保护



Baase (2013), A gift of fire: social, legal, and ethical issues of computing technology, pp. 55-56.



(A) Intentional Use or Release

这一类风险的重点是：用户自己主动（intentionally）公开了信息，但这种公开行为本身就可能带来隐私威胁。



为什么这属于 A 类“有意发布”？

因为：

所有信息都是“用户自愿发布的（intentional）”；

但用户其实没有充分意识到风险；

平台通过设计鼓励他们“过度公开”；

最终造成了严重隐私威胁。

所以这是“自愿但危险的公开”。

Hey, do you have a Twitter account? Have you ever noticed those messages in which people tell you where they are? Pretty annoying, eh_ Well, they're actually also potentially pretty dangerous. We're about to tell you why.

Don't get us wrong, we love the whole location-aware thing. The information is very interesting and can be used to create some pretty awesome applications. However, the way in which people are stimulated to participate in sharing this information, is less awesome. Services like Foursquare allow you to fulfill some primeval urge to colonize the planet. A part of that is letting everyone know you own that specific spot... You get to tell where you are and if you're there first, it's yours.

(website no longer available)





无意泄露或粗心丢失

(C) Inadvertent Leakage or Careless Loss

这一类隐私风险指的是：并非用户或公司“故意”公开数据，而是因为疏忽、软件漏洞、系统安全薄弱、配置错误等 导致数据意外泄露。

The Hack

Yahoo chief information security officer Bob Lord wrote in a statement on Yahoo's Tumblr site that the company had been the victim of a hacker intrusion in late 2014 that accessed at least 500 million accounts and retrieved a bounty of information, including user names, email addresses, telephone numbers, dates of birth, security questions and answers, and passwords—albeit passwords protected by cryptographic hashing. "We have confirmed that a copy of certain user account information was stolen from the company's network in late 2014 by what it believes is a state-sponsored actor," Lord writes. "An increasingly connected world has come with increasingly sophisticated threats. Industry, government and users are constantly in the crosshairs of adversaries."

Earlier Thursday Recode reported that Yahoo was expected to confirm a data breach that affects hundreds of millions of users. The site referenced a collection of 200 million of Yahoo's user names, birthdates, email addresses and hashed passwords that's been offered for sale on the dark web marketplace The Real Deal since at least August. In June, WIRED interviewed the hacker known as Peace or Peace of Mind, who's behind the data sale on Real Deal. Peace claimed to be a former member of a team of Russian cybercriminal hackers. He or she later sent WIRED a sample of the purported Yahoo data, but when WIRED sent test messages to the email addresses, half of them were invalid.

But Yahoo's announcement suggests a different breach. The timing, scale and Yahoo's claim of state involvement indicate it may be distinct from the one that surfaced data on the dark web and could also be significantly more serious.

WIRED

HACK BRIEF: YAHOO BREACH

HITS HALF A BILLION USERS



<https://www.wired.com/2016/09/hack-brief-yahoo-looks-set-confirm-big-old-data-breach/>

1. 无意泄露 (Inadvertent Leakage) = 非故意 + 非授权 + 安全不力

并不是有人“想”泄露，而是：

- 系统太弱
- 密码加密方式落后
- 安全更新没做
- 攻击者利用漏洞
- 公司监测不够

导致 不小心就泄露了大量用户数据。

2. “数据存在即风险” — The existence of data presents a risk

只要数据被保存：

- 就可能被黑
- 就可能被盗
- 就可能被出售
- 就可能被国家/组织要求交出

因此存储越多 → 风险越大。

(这是 Raase 书里重点句子)

3. 泄露规模越来越大 = 现代隐私最大威胁之一

Yahoo 5 亿用户数据泄露：

- 不是个人层面的问题
- 是全球级别、系统级别隐私威胁

现代数据泄露往往：

- 持续数年
- 公司直到很晚才发现
- 用户完全不知道自己的数据正在被黑客利用

4. “去加密密码” (hashed passwords) 也不是完全安全

因为：

- 弱哈希算法 (如 SHA1) 容易被破解
- 用户常用同一个密码
- 密保问题 & 出生日期可以用来重置密码

5. 数据一旦进入暗网 = 永远无法找回

黑客出售的数据可被：

- 复制无限次
- 广泛传播
- 被不同犯罪集团反复利用

属于永久性隐私损害。

 C 类风险一句话总结 (考试可直接写)

C = 因为疏忽、安全不足或意外导致的数据泄露 (包括黑客入侵、配置错误、丢失设备等)。
数据一旦泄露 → 不可逆扩散 → 用户无法挽回。



(E) Re-identification of Individuals

Re-identification (重新识别) 指的是：即使一个数据集已经“去标识 (anonymized)”，黑客或研究者仍然可以通过其他信息把具体个人“拼出来”。这一类隐私风险非常重要，因为很多系统（包括政府、公司和科研）都以为：“我们删了姓名/ID 号，就没有隐私问题了。”然而这其实非常不安全。

- Data can be anonymized by removing directly identifiable fields such as name, personal ID, or mobile phone number.

数据集通常通过删除直接识别字段 (direct identifiers) 来进行“匿名化”，例如：name (姓名) personal ID (身份证号) mobile phone number (手机号)

- However, often individuals may be re-identifiable by secondary data such as postal code (zip code), date of birth, and gender.

<https://www.isaca.org/resources/news-and-trends/industry-news/2024/reidentifying-the-anonymized-ethical-hacking-challenges-in-ai-data-training>

即使名字删掉了，很多“二级数据 (secondary data)”依然能识别你：postal code (邮编) date of birth (出生日期) gender (性别) 这三个加起来 在美国可以唯一识别 87% 的人 (著名研究)。

- Indeed, for individuals with infrequent characteristics, they may be identified even from other characteristics by linking data sets.

<https://www.fuqua.duke.edu/duke-fuqua-insights/jiaming-xu-does-data-anonymization-really-hide-your-identity>

如果你有一些“少见特征” (infrequent characteristics)，你更容易被识别

- Therefore, the anonymization process requires careful consideration.

真正的 re-identification 常发生在“多个数据集交叉链接 (linking datasets)”。



(F) Smart Phone Data

这一类隐私风险强调的是：现代智能手机与车载系统会自动收集、上传大量行为数据。这些数据比你想象得敏感得多。

Report: Porsche Opts for CarPlay Over Android Auto (October 2015)



Porsche has reportedly opted for Apple's rival CarPlay system over Android Auto due to privacy concerns.

- Porsche's 2017 911 sports car will have a number of new features, but Android Auto won't be one of them.
- That's because automakers that sign up to install Google's car infotainment system are required to collect and send back certain data to the company -- such as a vehicle's speed, RPMs and coolant temperatures, according to Motor Trend... That's information that Porsche isn't keen on sharing. In comparison, Apple CarPlay only checks to see whether a vehicle is in motion.

<https://www.pcmag.com/news/report-porsche-opts-for-carplay-over-android-auto>.



(G) Massive Data Storage (the Cloud)

云端大量存储数据带来的隐私风险，尤其是跨境传输（cross-border data transfer）的监管问题。

The screenshot shows the DarkReading website interface. At the top, the logo 'InformationWeek DARKReading' is displayed with the tagline 'CONNECTING THE INFORMATION SECURITY COMMUNITY'. Below the logo is a navigation bar with links: Home, News & Commentary, Authors, Slideshows, Video, Radio, Reports, White Papers, and Events. A secondary navigation bar lists various security topics: ANALYTICS, ATTACKS / BREACHES, APP SEC, CAREERS & PEOPLE, CLOUD, ENDPOINT, IoT, MOBILE, and OPERATIONS. The main content area features a 'RISK' section header. Below it, the article title 'What The EU's Safe Harbor Ruling Means For Data Privacy In The Cloud' is prominently displayed. The article is dated '10/6/2015 04:05 PM' and is authored by 'Michael Fey', with a 'Commentary' tag. The article text begins with 'The European Court of Justice today struck down the 15-year-old data transfer agreement between the European Union and the US. Here's how to begin to prepare for the fallout.' It continues to discuss the 'Snowden effect' and the impact of the ruling on U.S. businesses. On the left side of the article, there are social media sharing icons for LinkedIn, RSS, and Facebook, along with a 'Connect Directly' link. Below these, there is a '0 COMMENTS COMMENT NOW' section. At the bottom left, there is a 'Login' button and a 'Like 79' button, with a 'Tweet' button and a 'Share' button below it. A small progress indicator shows '50%'.

<http://www.darkreading.com/risk/what-the-eus-safe-harbor-ruling-means-for-data-privacy-in-the-cloud/a/d-id/1322512>

October 2015



Privacy Risks: Homework

Find other examples of privacy risks under each category A to K.



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Theories of Privacy

What do we mean by privacy?
Warren & Brandeis's argument
Solove's Taxonomy



Definitions of Privacy

- Three dictionary definitions of privacy:
 1. Seclusion (*being set apart, or out of view*) 隔离/不在视线中
 2. Secrecy or concealment 秘密/隐藏
 3. Freedom from intrusion 免于侵扰的自由

Scenario: You post your phone number on an online forum and shortly afterwards you start to receive harassing calls.

- Question: Harassing calls are immoral, but do they violate your privacy?
 - If privacy is seclusion or secrecy, then perhaps not - it was your responsibility to keep your phone number secret.
 - If privacy is freedom from intrusion, then harassing calls are invading your privacy.



The Right to Privacy

“Recent inventions and business methods call attention to the next step which must be taken for the protection of the person, and for securing to the individual... the right “to be let alone”. Instantaneous photographs and newspaper enterprise have invaded the sacred precincts of private and domestic life; and numerous mechanical devices threaten to make good the prediction that “what is whispered in the closet shall be proclaimed from the house tops.”

The Right to Privacy. Samuel. V. Warren and Louis D. Brandeis.
Harvard Law Review, Vol. 4. No. 5, (Dec. 15, 1890), pp. 193-220.

<https://www.jstor.org/stable/1321160>



The Right to Privacy

UN Declaration of Human Rights



Article 12: Right to privacy

You have the right to protection if someone tried to harm your good name, enter your home without permission or interfere with your correspondence.

No one shall be subjected to arbitrary interference with his privacy, family, home or correspondence, nor to attacks upon his honour and reputation. Everyone has the right to the protection of the law against such interference or attacks.

UNDHR is signed by most countries including China, USA and UK.

<https://www.ohchr.org/en/universal-declaration-of-human-rights/illustrated-universal-declaration-human-rights>



Warren & Brandeis's argument

反对者认为：隐私不是一种独立权利，只是其他权利的副产品。

- Opponents state that “privacy” derives from other rights:

- **Implicit contract:** 隐含契约

e.g., a department store that video records you in the changing rooms is violating an implicit contract with you not to record, rather than violating a separate right to privacy.

- **Intellectual property:** 知识产权

e.g., your friend posts an embarrassing picture of you on WeChat and tags you – this violates your intellectual property rights to your pictures of yourself and not a violation of your privacy.

- Warren & Brandeis debunked this; i.e., “Privacy” is not protected by contracts and intellectual property.

W&B 认为这些观点非常不够，他们提出：隐私不是知识产权，也不是契约问题。隐私必须作为一种独立权利存在。

为什么？因为隐私问题中有许多情形不属于契约或知识产权范畴。

- Using counter-argument of private letters published against wishes of the author to another person. The private letters are not covered by contract or intellectual property protection.



Solove's Taxonomy of Privacy

“Using the traditional method – seeking to define privacy’s essence or core characteristics ... [results in] endless disputes over what falls inside or outside the domain of privacy”

"I've Got Nothing to Hide" and Other Misunderstandings of Privacy – Daniel J. Solove

Bottom Up approach:

- Impossible to adequately define “privacy” in a “top-down” way that covers all instances.
- Identify instances of clear violations of privacy, and build up from there: *a hierarchical Taxonomy* of Privacy.



Solove's Taxonomy of Privacy

Four categories of privacy violation:

- **Information collection** 信息收集
- **Information processing** 信息处理
- **Information dissemination** 信息传播
- **Invasion** 直接侵入

Let's look at each in turn.



Information Collection

- **Surveillance:** Monitoring continuously, usually via audio, visual, or computer technology. 监视：持续性地监控，通常通过音频、视频或计算机技术。
- **Interrogation:** “Pressuring... individuals to divulge information”

盘问/诱导/欺骗获取信息施压、引诱人们透露信息

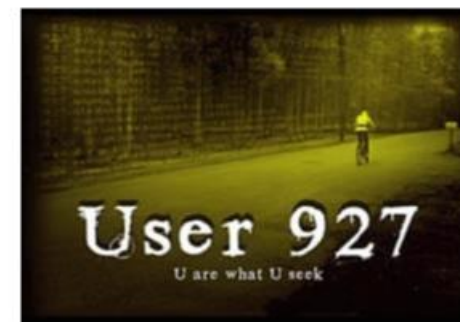


The 2017 model of the Porsche 911 will come with Apple CarPlay support. Photo: Simon Maina/Getty Images



Information Processing

- **Aggregation:** Collecting many small pieces of information about a person and linking them together, to create new information. 聚合：收集零散的信息，把它们拼在一起，从而构建“新信息”（特别是一个人的完整画像）。
- **Identification:** “Connecting information to individuals”. 识别：把数据与某一个具体的人绑定起来。
- **Insecurity:** Inadequately safeguarding collections of personal data against theft. 不安全/数据保护失败：没有妥善保护数据，导致数据可能被盗、泄露。
- **Secondary Use:** Using data that people willingly gave for one purpose for some other purpose they did not approve. 二次利用：用户为某个目的提供数据，却被公司用于其他未经同意的目的。
- **Exclusion:** Failing to notify individuals that their data is being collected, or failing to provide a way for individuals to view or correct such data. 排除：未告知用户数据被收集；或不给用户查看 / 修改数据的权利。





Information Dissemination

信息传播 (Information Dissemination) 大意：这类隐私侵犯都属于：个人信息本来已经被收集，但随后被不当共享、公开、传播或利用，造成伤害。重点是：数据 被传出去/被展示出去。

- **Breach of Confidentiality**: Breaking a contractual ... duty to keep someone else's information private. 违反保密义务：违背“保密义务”泄露别人的信息。

- **Disclosure**: Publishing private, but true, information in a way that damages the reputation of the subject. 未经同意的披露：公开“真实的私人信息”，并造成名誉损害。

- **Exposure**: Publicly displaying certain physical or emotional attributes of another that are normally considered private, especially if such as display is humiliating or embarrassing. 暴露：公开展示“身体或情绪上应当是私密”的内容，尤其是让人羞辱的场景。

- **Increased Accessibility**: Making records that are technically available to the public easier to access. 提高可访问性：信息本来不是完全私密，但：很难找到很不容易访问

但你把它整理、总结、搜索化、放网上，使其“容易访问”了 构成隐私侵犯。

- **Blackmail**: Exerting power over another by threatening to reveal damaging information about that person. 勒索：用“隐私”威胁他人：不给钱就曝光不答应某事就公开照片用聊天记录威胁对方

- **Appropriation**: Using someone else's identity for one's own ends. 冒用身份：冒用关键身份信息以获取利益。

- **Distortion**: “Manipulat[ing] ... the way a person is perceived or judged by others”. 委屈/捏造：通过编辑、伪造、断章取义让别人对你产生错误认知。



mugshots.com



Invasion

“Invasion” 不是曝光信息，也不是数据泄露，而是：直接干扰个人生活、心理空间或决策自由的行为。

侵扰：以让别人不舒服、扰乱宁静的方式与其沟通或接触。重点在于：打扰（disturbance）和 不受欢迎的干涉（unwanted contact）。

- **Intrusion**: Communicating with people in a way that disturbs their peace or makes them feel uncomfortable.

➤ E.g. **Phishing**: a cybercrime that uses **fake emails, websites, or messages** to trick people into sharing personal or financial data. **网络钓鱼**：虚假邮件，网站或者信息来诱导分享个人或财政数据

干涉个人决策：侵入或影响个人在私人事务上的选择权。它意味着：不是泄露你的信息，而是操纵你的行为。

- **Decisional Interference**: Intruding into an individual's decision making regarding their private affairs.

隐私不仅是“别人不要看我的东西”，
还包括：别人不应该操控我做决定。
也就是：
个人自主权（autonomy）被干扰。

➤ Mobile game Pokemon Go tracks user locations and “leads” players to establishments of businesses that paid the company to lead the players there.

https://privacy.wiki/Pokemon_Go_Mobile_Game



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Different Opinions on Privacy



Different Opinions on Privacy

- The “***Nothing to Hide***” argument: Richard Posner believes privacy as a social good is greatly overrated.

<http://bigthink.com/videos/judge-richard-posner-privacy>

- Juan Enriquez on “*Your Online Life, Permanent as a Tattoo*”

https://www.ted.com/talks/juan_enriquez_your_online_life_permanent_as_a_tattoo

- Alessandro Acquisti on “*What Will a Future Without Secrets Look Like?*”

http://https://www.ted.com/talks/alessandro_acquisti_what_will_a_future_without_secrets_look_like

- Glenn Greenwald on “*Why Privacy Matters?*”

▪http://https://www.ted.com/talks/glenn_greenwald_why_privacy_matters

思想家	观点一句话总结	立场
Posner	隐私被高估，透明更好	隐私不重要
Enriquez	线上行为永久，隐私是“数字纹身”	隐私重要
Acquisti	AI 能推断任何秘密，没有秘密的世界危险	强烈支持隐私保护
Greenwald	隐私是自由与民主的基础，“无可隐藏”是谬论	隐私至关重要

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